

# Golden Thread Plugin

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## Golden Thread Plugin — Documentation

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### Version gt 0.9.13 / gt-wiki 0.1.1

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Golden Thread turns an Obsidian vault into the single source of truth for all AI memory across every project and every session. The tiered rule model introduced in v0.6.0 now carries **seven hook-backed Core rules** enforced at three points in the turn, 0.9.12 added `gt-route` for the middle of a session, and 0.9.13 makes the session-start component check verify that the hooks are **wired**, not merely installed. gt-wiki 0.1.1 provides an LLM-powered knowledge base with immutable sources and interlinked pages.

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### What It Does

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Instead of scattered `.claude/memory/` files and CLAUDE.md snippets that live and die per session, Golden Thread gives every fact a permanent home in a structured vault. Knowledge flows up a hierarchy from session notes into the vault, and the right facts are always in scope when you need them.

```
session conversation
  ↓ gt-work / gt-promote
project memory (memory/*.md)
  ↓ gt-promote
project files (research.md / decisions.md / design.md / spec.md / runbook.md)
  ↓ gt-promote
Knowledge/ (cross-project wiki pages, backed by immutable Sources/)
  ↓ gt-promote
global-memory/ (loaded in every session, all projects)
```

Facts move up the hierarchy as they prove themselves general. They never move back down and are never silently deleted.

## Core Rules (gt 0.9.13)

Golden Thread defines a tiered rule model that separates rules by scope and enforcement strength. **Seven Core rules ship as of 0.9.13**, up from one at 0.6.0:

| # | Rule                                                                                                                                                                         |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Register your session and claim a vault file before writing it; never write a file another live session has claimed.                                                         |
| 2 | Never put a secret's value into the session — not to inspect it, not to redact it, not to check it.                                                                          |
| 3 | Begin every response with the current wall-clock timestamp — before any other text you emit.                                                                                 |
| 4 | <code>global-memory/</code> contains only facts needed in EVERY project.                                                                                                     |
| 5 | Do not auto-load the full memory index.                                                                                                                                      |
| 6 | A secret's value rests only in the secrets store or a mode-600 file the store wrote — never in source, a vault file, a repo, a log, or a session.                            |
| 7 | Label every derived figure you present as fact with its verification state — <code>unverified</code> , <code>self-verified</code> , OR <code>independently verified</code> . |

Verify the set at any time with `echo '{}' | ~/.claude/golden-thread/hooks/inject_core_rules.sh` .

### Three scope levels:

| Level   | Meaning                                                            |
|---------|--------------------------------------------------------------------|
| Core    | Applies to every chat, every project, no exceptions. Un-removable. |
| Context | Applies to this project or session type only.                      |

| Level   | Meaning                       |
|---------|-------------------------------|
| Generic | Default; no special standing. |

**Two enforcement strengths:**

| Strength  | Meaning                                                                                   |
|-----------|-------------------------------------------------------------------------------------------|
| Reminder  | Injected on every turn via <code>UserPromptSubmit</code> hook.                            |
| Validated | Checked by a <code>stop</code> hook that blocks any reply violating it; machine-enforced. |

Since 0.9.5 a third point exists: a `PreToolUse` hook ( `guard_session_claims.sh` ) denies a `Write / Edit` to a vault file another live session holds a claim on. It fails open on any parse failure, and deliberately does not cover `Bash` writes — parsing arbitrary shell fails open so often it would train the guard to be ignored.

**Core/Validated rules are hook-backed.** Hook scripts install to `~/.claude/golden-thread/hooks/` during `install.sh`. `settings.json` references them by absolute path so they survive vault renames and project moves. `gt_paths.py` is a self-healing resolver — if a recorded path is stale, it locates `core-rules/` by scanning the vault rather than failing silently.

The canonical rule definitions live in `Projects/golden-thread/core-rules/` inside the vault. Editing a rule file there changes what the hook injects — the rule text is never duplicated into the script.

**The canary:** `core_timestamp_every_message.md` is designated Core/Validated not because a missing timestamp causes real harm, but because it is trivially observable at the top of every reply. When the timestamp disappears, enforcement has broken.

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## gt Skills (15)

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### Setup

| Command                  | What it does                                                                                             |
|--------------------------|----------------------------------------------------------------------------------------------------------|
| <code>/gt:gt-init</code> | Create a new vault from scratch, or wire an existing vault to a project.<br>Idempotent — safe to re-run. |

| Command                    | What it does                                                                                                                                                       |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>/gt:gt-create</code> | Scaffold a new project folder with the standard structure. Captures your brain dump into <code>idea.md</code> . Supports sub-projects, tags, and runbook creation. |

## Daily Work

| Command                    | What it does                                                                                                                                                                                                                                        |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>/gt:gt-open</code>   | Load a project at session start. Reads all project docs in order (idea → research → decisions → design → spec → runbook → memory), summarizes state, and asks where to pick up.                                                                     |
| <code>/gt:gt-route</code>  | Mid-session. Names what the session has actually become, says where its output belongs, and checks you are in the right project, harness and model. For when a session drifted from what it opened with, or you cannot name what you are doing.     |
| <code>/gt:gt-work</code>   | Write back session findings. Appends to <code>research.md</code> , adds ADRs to <code>decisions.md</code> , refines <code>design.md</code> , creates <code>spec.md</code> when design is complete, and flags content for <code>PROTOCOL.md</code> . |
| <code>/gt:gt-ingest</code> | Bulk-import an existing project's memory files, CLAUDE.md rules, and notes into the vault. External sources are stored immutably in <code>Sources/</code> before being synthesized into Knowledge pages.                                            |
| <code>/gt:gt-review</code> | Scan recent Obsidian daily notes for uncaptured tasks and ideas. Surfaces them grouped by date, then promotes selected ones into tracked project folders.                                                                                           |

## Knowledge Management

| Command                     | What it does                                                                                                                                 |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <code>/gt:gt-query</code>   | Look something up — reads <code>index.md</code> first, follows wikilinks, falls back to grep. The vault is always checked before the web.    |
| <code>/gt:gt-promote</code> | Graduate a fact up the hierarchy: project memory → project files → Knowledge wiki page → global-memory. Also handles new project scaffolding |

| Command                     | What it does                                                                                                                                                                             |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | and retiring stale content.                                                                                                                                                              |
| <code>/gt:gt-refresh</code> | Check <code>Sources/</code> for upstream changes. Supersedes outdated sources with new immutable files — never edits the old one. Updates Knowledge pages that cited the changed source. |

## Context & Verification

| Command                      | What it does                                                                                                                                                                                                                               |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>/gt:gt-farm</code>     | Route bulk, mechanical, or second-opinion tasks to an external AI service as a self-contained work packet. All four gates (Stateless, Self-contained, Checkable, Releasable) must pass before a task leaves. Results come back unverified. |
| <code>/gt:gt-validate</code> | Verify a claim by re-deriving it with a fresh-context validator — never by reviewing the reasoning that produced it. Use before recording a finding as fact or before a production change.                                                 |

## Maintenance

| Command                          | What it does                                                                                                                                                                                                                                                                            |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>/gt:gt-lint</code>         | Audit the vault for structural problems: 14 checks covering broken wikilinks, orphaned pages, missing index entries, unlisted memory files, Knowledge pages citing superseded sources, stale pages, and <code>core-unenforced</code> — a Core rule that is stored but wired to no hook. |
| <code>/gt:gt-runbook-lint</code> | Scan all project <code>runbook.md</code> files for content that has drifted into multiple runbooks. Routes duplicated content to the right shared layer via <code>gt-promote</code> .                                                                                                   |
| <code>/gt:gt-settings</code>     | View and change what Golden Thread does on its own: component drift checking at session start, and the session report card at compact. Every automatic behaviour can be switched off.                                                                                                   |

## Installed, and actually wired (0.9.13)

The session-start component check answers two questions, not one:

| Question                              | How                                                                                                                                                |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Are the right files here, unmodified? | every shipped file is hashed into <code>MANIFEST.json</code> at install time and compared                                                          |
| Is any of it connected to anything?   | <code>MANIFEST.json</code> also declares the nine hook entries the plugin expects in <code>settings.json</code> , and the check compares those too |

Before 0.9.13 only the first question was asked, so a machine with every file installed and an empty `settings.json` reported **components: clean** while nothing ran. The clean line now reads *"installed matches , all 9 hooks wired"* — the second clause is the part that was missing.

Two states are reported: `unwired` (no entry for that event names the script — it never runs) and `badpath` (wired, but the command names a path that does not exist on this machine, which is what a deleted version directory or an unquoted path containing a space produces).

The declaration lives in `gt_components.HOOK_REGISTRATIONS` and is what `install.sh` registers *from*, so the installer and the checker cannot disagree about what "wired" means. `install.sh` verifies its own six entries immediately after writing them; the three enforcement hooks are owned by `vault_init.py` `install-core-rules`, which needs a vault. `selftest.sh` asserts all nine from outside — the only vantage point that still works when nothing is wired at all.

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## gt-wiki Skills (5)

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The gt-wiki plugin provides an LLM-powered knowledge base separate from the Golden Thread memory hierarchy. Sources are ingested immutably; Knowledge pages are synthesized summaries linked bidirectionally.

**Key design decision:** The vault `CLAUDE.md` is kept slim (architecture + pointer only). Page format schema lives in `Knowledge/_template.md` and is read on demand by the ingest and refresh skills — it is not loaded into every session's context.

| Command                          | What it does                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>/gt:gt-wiki-init</code>    | Set up a new wiki vault from scratch. Runs <code>vault_init.py</code> deterministically — creates <code>Sources/</code> , <code>Knowledge/</code> , <code>index.md</code> , <code>log.md</code> , seeds <code>CLAUDE.md</code> from template, and stamps <code>Knowledge/_template.md</code> .                                                                                                                          |
| <code>/gt:gt-wiki</code>         | Query the wiki. Reads <code>index.md</code> first, follows wikilinks, falls back to grep, deep-digs Sources for precision. Logs every query.                                                                                                                                                                                                                                                                            |
| <code>/gt:gt-wiki-ingest</code>  | Add a source: fetch or paste, store immutably in <code>Sources/</code> , discuss with user, write Knowledge pages per <code>_template.md</code> , cross-link bidirectionally, then update <code>index.md</code> and <code>log.md</code> through <code>wiki_log.py</code> so every entry has the same shape. Records <code>upstream_sha:</code> for sources inside a git repo.                                           |
| <code>/gt:gt-wiki-lint</code>    | Run <code>wiki_lint.py</code> (10 deterministic checks). Interprets findings, proposes fixes, records declines in <code>lint-declines.md</code> so nothing gets re-litigated.                                                                                                                                                                                                                                           |
| <code>/gt:gt-wiki-refresh</code> | Check selected sources for upstream changes. <code>wiki_refresh.py</code> detects changes deterministically for local sources ( <code>git fetch</code> + <code>git diff</code> from the source's <code>upstream_sha:</code> , or its <code>ingested:</code> date) and flags web-only sources for the LLM to fetch and compare. Supersedes changed ones with new immutable source files. Updates citing Knowledge pages. |

## Vault Structure

```
<vault>/
  CLAUDE.md           ← architecture + pointer to Knowledge/_template.md
  index.md            ← navigational index of all Knowledge pages
  log.md              ← audit trail: every create, ingest, promote, retire
  review-queue.md     ← items flagged for owner review (written by gt-lint)

  Sources/            ← IMMUTABLE raw originals
    YYYY-MM-DD <title>.md

  Knowledge/          ← cross-project wiki pages
    _template.md      ← page format schema (read by ingest/refresh skills)
    <Page Title>.md
```

```

global-memory/
  MEMORY.md          ← index; loaded every session
  <topic>.md

Projects/
  README.md          ← master project list
  CONVENTIONS.md     ← lifecycle phases, file roles, tag taxonomy
  PROTOCOL.md        ← cross-project process rules
  INFRASTRUCTURE.md  ← server fleet, defined once

<project-slug>/
  README.md          ← status board + YAML frontmatter
  source.md          ← where code lives, topology, deploy file plan
  idea.md            ← original brain dump – IMMUTABLE
  research.md        ← append-only findings
  decisions.md       ← append-only ADRs
  design.md          ← current architecture
  spec.md            ← handoff artifact (created when design is complete)
  runbook.md         ← operational procedures (optional)
  memory/
    MEMORY.md        ← session memory index
    <topic>.md

golden-thread/
  core-rules/        ← canonical Core rule definitions

```

## Immutability Model

| File         | Mutability  | Rule                                           |
|--------------|-------------|------------------------------------------------|
| Sources/*.md | Immutable   | Never edited; superseded by new files          |
| idea.md      | Immutable   | Origin story; never changed                    |
| research.md  | Append-only | History is extended, never rewritten           |
| decisions.md | Append-only | Reversed by adding a new ADR, never by editing |
| design.md    | Mutable     | Always describes the current architecture      |

| File                         | Mutability           | Rule                                              |
|------------------------------|----------------------|---------------------------------------------------|
| <code>spec.md</code>         | Mutable (scope only) | Self-contained handoff doc                        |
| <code>runbook.md</code>      | Mutable              | Operational HOW-TO; incubator before CLAUDE.md    |
| <code>PROTOCOL.md</code>     | Append-only          | Cross-project process rules                       |
| <code>Knowledge/*.md</code>  | Mutable              | Synthesized summaries; cite <code>Sources/</code> |
| <code>core-rules/*.md</code> | Mutable              | Editing here changes what hooks inject            |

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## Context Footprint

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| What loads                                                    | When                               | Approx tokens |
|---------------------------------------------------------------|------------------------------------|---------------|
| <code>~/.claude/CLAUDE.md</code> (global instructions)        | Every session                      | ~200          |
| <code>&lt;vault&gt;/CLAUDE.md</code> (architecture + pointer) | Every session in vault dir         | ~100          |
| <code>global-memory/MEMORY.md</code> (index only)             | Every session                      | ~50           |
| <code>Knowledge/_template.md</code> (page schema)             | Only during ingest/refresh         | ~150          |
| Project memory files                                          | On demand via <code>gt-open</code> | Varies        |

The slim CLAUDE.md design (introduced in gt-wiki 0.1.0) drops the vault's cold-start footprint from ~1,300 tokens to ~350.

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## Source Topology

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Every project records where its code lives in `source.md` .

| Topology                    | Meaning                                                |
|-----------------------------|--------------------------------------------------------|
| <code>local</code>          | A folder on this machine. No deploy step.              |
| <code>remote</code>         | One repo, one server; server pulls from GitHub.        |
| <code>bastion-jump</code>   | Several servers, all reached through one gateway host. |
| <code>bastion-direct</code> | Several servers, each reachable independently.         |

Bastion projects carry a **file plan** marking each deployed file `static` (byte-identical everywhere) or `unique` (per-server variant that must never be cross-deployed).

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## Install

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```
# From zip
unzip golden-thread-plugin.zip
cd golden-thread-plugin
bash install.sh
# Restart Claude Code
```

Installs both `gt` (v0.9.13) and `gt-wiki` (v0.1.1) as separate plugins under the `golden-thread-plugin` marketplace. Requires Python 3.8+.

`install.sh` installs the **newest version directory** present, not a hardcoded constant — pass an argument only to roll back deliberately ( `./install.sh 0.9.3` ). Never pipe it to `head` : `set -o pipefail` turns the closed pipe into an abort partway through, leaving the cache updated and registration undone.

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## Typical Session Pattern

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```
/gt:gt-open my-project          # load project, summarize state

# ... work happens ...
```

```
/gt:gt-route          # "where is this going?" – run it when the session
                      # has drifted, or before writing something down and
                      # you are unsure which file it belongs in

/gt:gt-work           # write back findings, update docs

# periodically

/gt:gt-lint           # catch structural drift
/gt:gt-wiki-ingest <url> # add a source to the wiki
/gt:gt-promote        # graduate a finding to Knowledge or global-memory
```

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## Requirements

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- Python 3.8+
- Claude Code (any version)
- Obsidian (optional — the vault is plain markdown; Obsidian is the GUI layer)